A Cloud Detection Method for Remote Sensing Images Combining Optical Images and SAR Images

Chengjie Hu, Bo Wang, Qinghong Sheng, Jun Li, Xiao Ling, and Xiang Liu

***Abstract*—Accurate cloud detection in optical (OPT) remote sensing imagery remains a persistent challenge, particularly for semi-transparent thin clouds and high-reflectance terrestrial surfaces where spectral signatures often overlap. This article proposes SAR-OPT-cloud detection network (CDNet), a novel multimodal fusion framework that synergistically integrates optical spectral information with the cloud-penetrating structural cues from synthetic aperture radar (SAR). To effectively bridge the modality gap, we develop a multimodal feature fusion module (MFFM) that explicitly disentangles and aggregates joint representations and cross-modal differential features under a channel-spatial attention mechanism. Furthermore, we introduce Mamba-inspired visual state space (VSS) blocks to facilitate efficient multidirectional contextual modeling with linear computational complexity, significantly enhancing the delineation of irregular cloud boundaries and the detection of fragmented cloud structures. Experimental results on the newly established Nanjing University of Aeronautics and Astronautics (NUAA)-SAR-OPT-cloud detection (CD) dataset—comprising temporally paired Sentinel-1 and Sentinel-2 imagery—demonstrate that our method achieves a state-of-the-art overall accuracy (OA) of 99.13% and an F1-score of 0.9178. Qualitative and quantitative analyses confirm that the proposed framework substantially reduces omission errors in thin-cloud regions and suppresses false alarms over high-reflectance surfaces, offering a robust and efficient solution for complex remote sensing applications.**

***Index Terms*—Cloud detection, deep learning, multimodal data, remote sensing image processing, state space model.**

I. Introduction

SATELLITE optical (OPT) imagery plays a fundamental role in environmental monitoring, land-cover mapping, and a wide range of geoscience applications. However, cloud contamination remains one of the primary factors limiting the effective utilization of optical observations, as clouds obscure surface information and alter spectral responses, thereby degrading the reliability of downstream analysis and retrieval tasks [1]. Accurately detecting clouds—especially thin clouds, cloud boundaries, and clouds over high-reflectance surfaces, such as snow, barren land, and urban areas—continues to be a challenging problem in optical remote sensing.

Traditional cloud detection approaches are mainly based on physical rules and thresholding strategies, including radiometric tests, spectral indices, and decision-tree schemes, such as automated cloud-cover assessment (ACCA) and Fmask [1], [9]. These methods exploit general physical characteristics of clouds, such as high visible reflectance and low thermal brightness temperature, and are widely used in operational processing chains [2], [12]. However, their performance deteriorates significantly in complex scenarios. Thin clouds often exhibit weak spectral signatures and mixed pixels, leading to omission errors, while clouds over high-reflectance surfaces are frequently misclassified due to spectral similarity, resulting in false alarms [3], [8].

To alleviate these limitations, temporal-difference-based methods incorporate multitemporal information to distinguish transient clouds from stable land surfaces [2], [13]. Although such strategies can improve robustness under certain conditions, they rely on dense time series and cloud-free reference observations, which are often unavailable in persistently cloudy regions. Machine-learning-based methods, including support vector machines and random forests, further improve cloud detection by learning decision boundaries from handcrafted spectral–textural features [14], [15], [16], [17]. Nevertheless, these approaches remain constrained by manually designed features and limited generalization capability under heterogeneous surface conditions.

In recent years, deep learning has substantially advanced cloud detection performance. Encoder–decoder architectures and fully convolutional networks, such as Cloud-Net, RS-Net, UNet variants, and DeepLab-based models, can learn hierarchical semantic representations directly from optical imagery and achieve state-of-the-art accuracy on benchmark datasets [6], [7], [18], [19], [20], [21], [22], [23], [24]. Boundary-aware and attention-enhanced networks further improve cloud edge delineation and segmentation quality [23], [24]. Despite these advances, optical-only deep models still struggle in scenarios where cloud signatures are weak or spectrally ambiguous. Thin clouds, cloud boundaries, and clouds over high-reflectance surfaces remain persistent failure cases due to the inherent limitations of optical sensing [5], [8].

A fundamental reason for these failures is that optical sensors cannot penetrate cloud layers, making it difficult to reliably separate cloud and surface signals when their spectral responses overlap. This limitation motivates the integration of

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complementary sensing modalities. Synthetic aperture radar (SAR) imagery provides cloud-penetrating capability and captures surface structure and dielectric properties that are largely insensitive to illumination and atmospheric conditions [26]. Previous studies have demonstrated that optical–SAR fusion can enhance performance in cloud removal, land-cover mapping, and semantic segmentation tasks by improving feature separability in spectrally ambiguous regions [4], [5], [29].

However, the potential of combining SAR and optical imagery for pixel-level cloud detection has not been fully explored. Most existing multimodal approaches focus on cloud removal or downstream interpretation tasks and typically rely on simple feature concatenation or generic attention mechanisms [4], [5]. These strategies do not explicitly distinguish between shared and differential cross-modal information, NOR do they systematically analyze how SAR contributes to reducing false detections over high-reflectance surfaces or improving thin-cloud recognition.

Motivated by these limitations, this study proposes SAR-OPT-cloud detection network (CDNet), a multimodal cloud detection framework that jointly exploits optical spectral information and SAR backscattering characteristics. The proposed network is designed to address three key challenges:

1) Reliable detection of thin and semi-transparent clouds;
2) Suppression of false detections over high-reflectance surfaces;
3) Accurate delineation of cloud boundaries under complex surface conditions.

To this end, SAR-OPT-CDNet adopts a dual-branch encoder architecture for modality-specific feature extraction, a multimodal feature fusion module (MFFM) that explicitly decomposes joint and differential cross-modal features, and an Mamba-inspired visual state space (VSS) Block for efficient multidirectional contextual modeling. By integrating complementary surface information from SAR with cloud-sensitive optical features at the representation level, the proposed framework is inherently robust to spectral ambiguity and minor temporal inconsistencies between modalities.

In addition, we construct a new multimodal benchmark dataset, Nanjing University of Aeronautics and Astronautics (NUAA)-SAR-OPT-cloud detection (CD), consisting of temporally paired Sentinel-1 and Sentinel-2 imagery covering ten representative land-cover categories, including snow/ice, barren land, urban areas, forests, croplands, and water bodies. The dataset emphasizes challenging scenarios, such as thin clouds, cloud boundaries, and clouds over high-reflectance surfaces, providing a focused testbed for evaluating multimodal cloud detection methods.

The main contributions of this work are summarized as follows.

1) A task-oriented optical–SAR cloud detection framework that improves cloud–surface separability in challenging scenarios.
2) A difference-aware multimodal fusion module that explicitly exploits cross-modal discrepancies to suppress false alarms and missed detections.
3) An efficient VSS-based contextual modeling strategy that enhances thin-cloud and boundary representation with moderate computational overhead.
4) A new multimodal cloud detection dataset facilitating systematic evaluation of optical–SAR fusion strategies.

II. Related Work

Related work is organized into four categories: 1) physical rule and temporal-difference methods; 2) machine-learning-based approaches; 3) deep learning for cloud and cloud–snow detection; and 4) multimodal optical–SAR fusion. For each class, we highlight the limitations that motivate the design choices of SAR-OPT-CDNet.

A. Physical Rule and Temporal-Difference Methods

Physical rule-based methods detect clouds using radiometric thresholds on visible reflectance, thermal brightness temperature, and spectral indices. Representative algorithms, such as ACCA, Fmask, and their variants [1], [2], [9], [10], [11], [12] are widely used in operational processing chains. They exploit generic physical properties of clouds—high reflectance in visible bands and low brightness temperatures in thermal bands—but struggle with thin clouds and clouds over high-reflectance surfaces (e.g., snow, ice, bright urban materials), where spectral signatures become ambiguous [3], [5], [8]. Temporal methods like Tmask and multitemporal cloud masking [2], [13] introduce time series information to distinguish transient clouds from stable land surfaces; however, they require dense acquisitions and cloud-free reference images, which are not always available, particularly in regions with persistent cloud cover.

In our work, these limitations are addressed by incorporating SAR as an auxiliary modality: instead of relying solely on optical radiometric thresholds or temporal contrasts, SAR-OPT-CDNet learns to jointly interpret cloud-affected optical signals and cloud-penetrating SAR backscatter, thereby reducing false detections in conditions where physical rule-based methods typically fail.

B. Machine Learning-Based Cloud Detection

Machine-learning approaches employ supervised classifiers, such as SVMs, random forests, and shallow neural networks trained on handcrafted spectral, textural, and morphological features [14], [15], [16], [17], [20], [21], [22]. These methods improve over pure thresholding by learning decision boundaries from labeled data. For instance, SVM- or PCANet-based models [16], [21] can better handle moderate variability in background surfaces, and elevation-assisted schemes further incorporate DEM information to discriminate clouds from snow [8]. Nevertheless, the feature space remains manually designed and often sensor-specific, limiting the robustness of these methods in heterogeneous landscapes and challenging cloud scenarios.

C. Deep Learning for Cloud and Cloud–Snow Detection

Deep learning has led to substantial improvements in cloud detection performance. FCN-based architectures such

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as Cloud-Net [7], UNet and its extensions [18], [23], [24], DeepLab-based models [19], and RS-Net [6] achieve accurate cloud masks by learning multiscale semantic features. Boundary-aware networks like BoundaryNet emphasize precise cloud edge refinement [23], and recent work using improved UNet3 + with attention modules further targets the joint detection of clouds and snow [8]. Moreover, larger benchmark datasets, such as CloudSEN12 [25] and related global collections support more robust training and evaluation of optical-based models.

Despite these advances, deep models operating solely on optical imagery share two inherent limitations.

1) *Spectral Ambiguity:* When clouds overlay high-reflectance surfaces (snow, bright sand, urban rooftops), optical signatures alone are insufficient to reliably distinguish clouds from underlying surfaces, leading to commission errors [3], [5], [8], [23].
2) *Weak Contrast and Boundary Uncertainty:* Thin clouds and cloud boundaries exhibit subtle spectral variations and mixed pixels, resulting in omission errors and blurred edges even for strong optical-based networks [5], [7], [8], [25].

Our SAR-OPT-CDNet directly targets these failures. First, it introduces SAR as an independent yet complementary modality that is insensitive to visible reflectance and thus helps resolve ambiguity over high-reflectance surfaces. Second, the VSS Block with SS2D scanning provides efficient multidirectional context modeling, which enhances the representation of thin structures and cloud boundaries. In Section V of this article, we show that these design choices yield improved F1 scores and precision specifically in thin-cloud and high-reflectance scenarios compared with strong optical-only baselines such as BoundaryNet.

D. Optical–SAR Fusion and Multimodal Remote Sensing

Optical–SAR fusion has attracted increasing attention in cloud removal, land-cover mapping, and semantic segmentation tasks [4], [5], [26], [29], [30], [31], [32], [33]. Meraner et al. [4] used SAR-guided deep residual networks to reconstruct cloud-free Sentinel-2 imagery, demonstrating that SAR backscatter can restore structural information obscured by dense clouds. Xu et al. [5] proposed GLF-CR, a global–local fusion framework that leverages SAR for cloud removal, achieving superior reconstruction quality in highly clouded regions. More broadly, multimodal fusion reviews and frameworks [29], [31], [32] highlight the potential of deep learning to integrate heterogeneous sources, such as SAR and optical imagery, as well as emerging cross-modal alignment methods for segmentation and classification [30], [31], [32], [33].

However, these multimodal efforts predominantly focus on cloud removal or downstream analysis, rather than explicit pixel-level cloud detection. Fusion strategies are typically based on early concatenation or simple attention mechanisms and seldom distinguish between shared and differential information across modalities [4], [5], [29]. Moreover, most works do not quantitatively analyze how SAR contributes to correcting specific cloud detection errors, such as thin-cloud omissions or bright-surface commissions.

Different from these studies, SAR-OPT-CDNet is designed as a cloud-detection-specific multimodal network with two distinctive characteristics.

1) *Task-Oriented Fusion With* MFFM: Instead of naïve concatenation, MFFM explicitly builds joint (shared) features to capture common structures and difference (differential) features to highlight discrepancies between optical and SAR responses. Channel and spatial attention further refine these signals to amplify discriminative cues for cloud versus surface separation.
2) Efficient *Long-Range Modeling With Mamba-Inspired VSS Block:* Inspired by recent state-space models and visual Mamba architectures [37], [38], [39], the VSS Block uses selective scanning to aggregate multidirectional context with lower computational cost than standard transformers. This design improves boundary localization and thin-cloud detection efficiency, which is validated in our ablation studies by consistent gains in F1 and precision with modest complexity overhead.

In summary, while existing optical-only and multimodal methods provide strong baselines, they either fall short in thin-cloud and high-reflectance scenarios or do not fully exploit cross-modal differences. SAR-OPT-CDNet is explicitly tailored to these challenging conditions and demonstrates, through both quantitative metrics and qualitative visual results, that carefully designed optical–SAR fusion and state-space-based context modeling can significantly reduce missed detections and false alarms in cloud detection.

III. Data

A. Sentinel-1 and Sentinel-2 Data

Sentinel-1, a critical Earth observation satellite in the European Space Agency's Copernicus program (GMES), consists of two satellites—Sentinel-1A and Sentinel-1B. It is equipped with a *C*-band SAR, which enables the acquisition of high-resolution radar imagery under any weather conditions, both during the day and at night. Sentinel-1 provides powerful surface observation capabilities, allowing effective monitoring of the Earth's surface even in cloud-covered or overcast conditions, offering a significant advantage in penetrating cloud layers. Sentinel-1 supports four imaging modes: stripmap (SM), interferometric wide (IW), extra wide (EW), and wave (WV), and provides four polarization combinations (VV, VH, HH, HV). These features make it particularly well-suited for surface observation, disaster monitoring, and marine monitoring. For this study, we used VV and VH polarization data from Sentinel-1 as input sources. VV polarization mainly captures surface reflection intensity, making it ideal for monitoring water bodies and surface changes, while VH polarization offers enhanced texture and structural information, crucial for distinguishing clouds from high-reflectance surfaces, such as ice, snow, and buildings. By combining VV and VH polarization data, the accuracy of cloud detection can be significantly enhanced. The Sentinel-1 band configuration used in this study is summarized in Table I.

Sentinel-2 is a high-resolution multispectral imaging satellite in the European Space Agency's Copernicus program,

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TABLE I
SENTINEL-1 BAND OVERVIEW

Band no.	Polarization	Frequency (GHz)	Wavelength (cm)	Spatial Resolution (m)	Mode
Band 1	VV	5.405	5.6	10	IW
Band 2	VH	5.405	5.6	10	SM
Band 3	HH	5.405	5.6	5	EW
Band 4	HV	5.405	5.6	40	WV

TABLE II
SENTINEL-2 BAND OVERVIEW

Band no.	Band name	Central Wavelength(μm)	Bandwidth(nm)	Spatial Resolution(m)
Band 1	Coastal Aerosol	0.443	27	60
Band 2	Blue	0.490	98	10
Band 3	Green	0.560	45	10
Band 4	Red	0.665	38	10
Band 5	Vegetation Red Edge	0.705	19	20
Band 6	Vegetation Red Edge	0.740	18	20
Band 7	Vegetation Red Edge	0.783	28	20
Band 8	NIR	0.842	145	10
Band 8A	Vegetation Red Edge	0.865	33	20
Band 9	Water Vapour	0.945	26	60
Band 10	SWIR-Cirrus	1.375	75	60
Band 11	SWIR	1.610	143	20
Band 12	SWIR	2.190	242	20

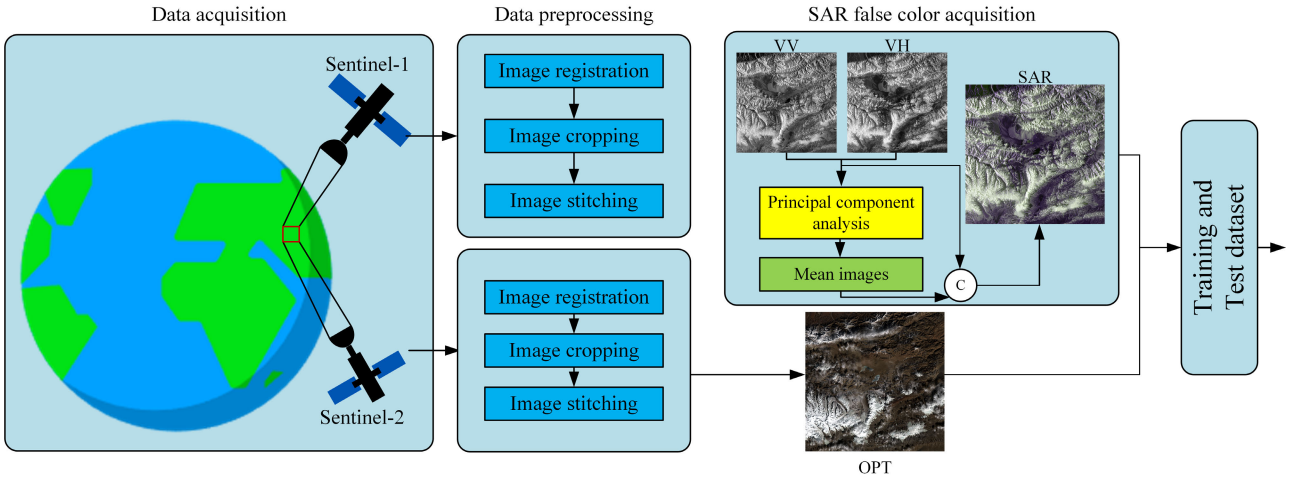


Fig. 1. Data Processing.

designed primarily for land observation. It carries a multispectral imager (MSI) that captures 13 spectral bands ranging from visible and near-infrared to shortwave infrared (wavelength range of 0.443–2.190 μm), with spatial resolutions of 10, 20, and 60 m, and an image swath width of up to 290 km. Sentinel-2's imaging capabilities cover a wide range of scenes, including vegetation, soil, water bodies, inland waterways, and coastal regions. The revisit cycle of a single Sentinel-2 satellite is ten days, but with two satellites working in complementary operation, the revisit period is reduced to five days, significantly enhancing temporal resolution. In this study, we primarily use the 2nd, 3rd, and 4th bands of Sentinel-2 (which correspond to blue, green, and red wavelengths) to generate high-resolution true-color visible images.

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TABLE III
DETAILS OF TRAINING AND TESTING IMAGES IN THE NUAA-SAR-OPT-CD DATASET

	No	Product ID	Main land cover type	Acquisition data
Train data	1	S2A_MSIL1C_20210126T052111_N0209_R062_T44SNE_20210126T063836	Snow/Ice/ Barren/water	2021-01-26
	2	S2A_MSIL1C_20180816T053641_N0206_R005_T44TKK_20180816T093424	Barren/Farmland/Water/Grass	2018-08-16
	3	S2A_MSIL1C_20190815T051651_N0208_R062_T44TPN_20190815T090034	Forest/Grass	2019-08-15
	4	S2A_MSIL1C_20190630T045701_N0207_R119_T45SWC_20190630T080543	Snow/Ice/Barren/Grass	2019-06-30
	5	S2A_MSIL1C_20180930T044701_N0206_R076_T45SXR_20180930T074413	Barren/Water	2018-09-30
	6	S2A_MSIL1C_20190714T043711_N0208_R033_T46TFN_20190714T073938	Barren/Snow/Ice/grass	2019-07-14
	7	S2A_MSIL1C_20201206T041141_N0209_R047_T47SMV_20201206T053320	Snow/Ice/ Barren/Water	2020-12-06
	8	S2A_MSIL1C_20191023T040821_N0208_R047_T47TQF_20191023T074550	Barren	2019-10-23
	9	S2A_MSIL1C_20200222T030731_N0209_R075_T49QEE_20200222T060244	Forest/Urban/Water/Barren	2020-02-22
	10	S2A_MSIL1C_20190928T031541_N0208_R118_T49RBJ_20190928T061248	Forest/Water/Urban/Grass	2019-09-28
	11	S2A_MSIL1C_20180722T030541_N0206_R075_T49RFP_20180722T060550	Farmland/Water/Urban/Grass	2018-07-22
	12	S2A_MSIL1C_20180429T032541_N0206_R018_T49SCV_20180429T062304	Forest/Barren/Water	2018-04-29
	13	S2A_MSIL1C_20191118T025011_N0208_R132_T50RMN_20191118T071843	Forest/Urban/Water	2019-11-18
	14	S2A_MSIL1C_20200317T024541_N0209_R132_T51TWM_20200317T053350	Barren/Water/Snow/Ice	2020-03-17
	15	S2A_MSIL1C_20200410T022551_N0209_R046_T51TXG_20200410T042047	Forest/Urban/Water/Grass	2020-04-10
	16	S2A_MSIL1C_20210207T023851_N0209_R089_T52UCU_20210207T040210	Tundra/Barren	2021-02-07
Test data	1	S2A_MSIL1C_20200528T050701_N0209_R019_T44SPC_20200528T082127	Barren/Water/Snow/Ice	2020-05-28
	2	S2A_MSIL1C_20191001T050701_N0208_R019_T45TXN_20191002T142939	Barren/Snow/Ice/Water	2019-10-02
	3	S2A_MSIL1C_20200416T042701_N0209_R133_T46SFE_20200416T074050	Barren/Snow/Ice/Water	2020-04-16
	4	S2A_MSIL1C_20190812T032541_N0208_R018_T48RXU_20190812T070322	Farmland/Urban/Water/Grass	2019-08-12
	5	S2A_MSIL1C_20180930T030541_N0206_R075_T49QDD_20180930T060706	Shrubland/Water/Urban/Grass	2018-09-30
	6	S2A_MSIL1C_20190328T033701_N0207_R061_T49TCF_20190328T071457	Barren/Water/Urban	2019-03-28

These bands provide valuable spectral information that assists in surface feature extraction and contributes high-quality optical input for cloud detection tasks. The Sentinel-2 band configuration used in this study is summarized in Table II.

B. Data Processing

To construct reliable multimodal samples for cloud detection, Sentinel-2 optical images were paired with Sentinel-1 SAR observations based on both temporal proximity and spatial consistency. Specifically, each Sentinel-2 image was matched with the closest Sentinel-1 acquisition within a ± 3 temporal window. This threshold was selected to optimize the trade-off between data availability and surface stability. As surface backscattering characteristics in the *C*-band typically remain stable over short intervals in the absence of extreme weather events, the structural information provided by SAR (e.g., topography and built structures) continues to offer a consistent reference for disambiguating clouds from high-reflectance surfaces. Consequently, this minor temporal offset does not compromise the network's ability to suppress false detections or recognize thin clouds."

All Sentinel-1 SAR data were obtained in IW mode with VV and VH polarizations and processed using the standard GEE preprocessing chain, including radiometric calibration, terrain correction, and geocoding. The SAR images were resampled to match the spatial resolution and grid of the corresponding Sentinel-2 imagery using bilinear interpolation. Owing to the high geolocation accuracy of both Sentinel-1 and Sentinel-2 Level-1 products, the residual co-registration error is within one pixel, which is sufficient for pixel-level cloud detection tasks. The overall data processing workflow is shown in Fig. 1.

It should be noted that SAR data are not affected by cloud cover and mainly capture surface structural and dielectric properties. Therefore, minor temporal offsets between optical and SAR acquisitions do not compromise cloud–surface discrimination, but instead provide complementary information to suppress false detections over high-reflectance surfaces and enhance thin-cloud recognition.

C. Nuaa-Sar-Opt-CD Dataset

In this study, we built a multimodal cloud detection dataset based on the WHUS2-CD+ cloud detection dataset, which contains well-matched SAR and optical data, with the cloud mask provided by WHUS2-CD+. The images selected span various typical land cover types, including farmland, forests, grasslands, shrublands, wetlands, water bodies, tundra, urban areas, barren lands, and snow/ice, covering ten primary categories. The dataset includes images with a time span from April 2018 to February 2021, encompassing all seasons and a diverse range of scenes and climatic conditions.

A total of 22 sets of Sentinel-1 and Sentinel-2A images were selected to construct the NUAA-SAR-OPT-CD dataset. Of these, 16 sets were used to build the training and validation sets, while six sets were allocated for testing. To meet the input requirements of the deep learning network, the training images were cropped into 512×512 pixel patches with a 50% overlap between adjacent patches. After cropping, a total of 26,554 image patches were generated, with 19,915 allocated for training and 6639 for validation. The detailed training and testing image list is provided in Table III.

For the test set, we specifically selected images from various land cover types to evaluate the algorithm's performance in complex surface scenarios. This includes 3 images featuring

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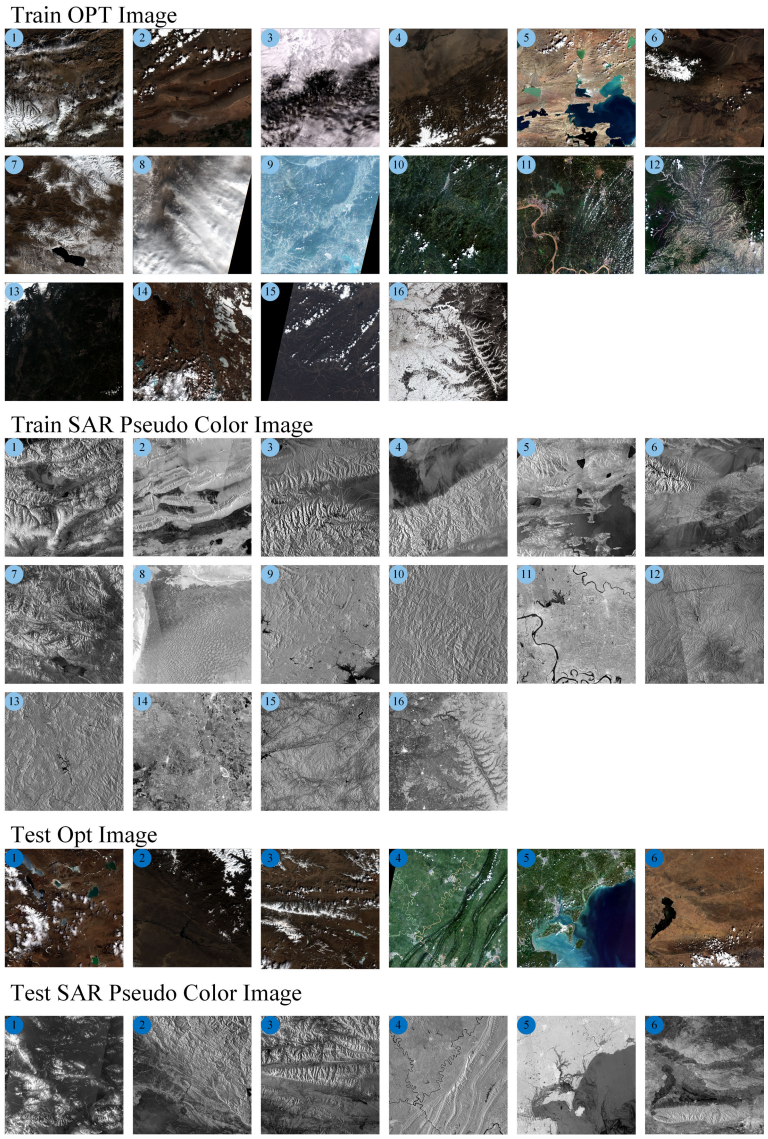


Fig. 2. Details of Training and Testing Images in the NUAA-SAR-OPT-CD dataset.

snow and ice, one image of barren land, and two images of urban areas. This dataset design carefully considers the challenges posed by thin clouds and clouds over highly reflective surfaces, ensuring a diverse and stringent set of validation conditions for the experiments. Representative training and testing images are shown in Fig. 2.

IV. Method

A. SAR-OPT-CDNet

SAR-OPT-CDNet is designed as a multimodal cloud detection framework that jointly leverages the spectral sensitivity of optical imagery and the cloud-penetrating structural information provided by SAR observations. The overall architecture follows an encoder–fusion–decoder paradigm, as shown in Fig. 3.

The model adopts a dual-branch encoder, where the optical and SAR streams independently extract modality-specific representations. This separation preserves complementary information: optical features emphasize spectral and texture patterns related to cloud thickness, while SAR features capture surface structure and scattering characteristics that help suppress false alarms over high-reflectance surfaces. After encoding, the two feature streams are integrated through the proposed MFFM, which combines joint representations and differential cross-modal cues under the guidance of channel–spatial attention. This design enables the network to emphasize modality-shared semantics while explicitly modeling discrepancies between SAR and optical signals around cloud edges and thin-cloud regions. To further strengthen long-range context modeling, the fused features are processed by the VSS Block, a lightweight state-space module inspired by Mamba. The VSS Block captures directional contextual dependencies with selective scanning, improving the delineation of cloud boundaries and the detection of thin and semi-transparent clouds.

Finally, the decoder aggregates multiscale fused features to generate pixel-level cloud masks. SAR-OPT-CDNet integrates complementary multimodal cues and efficient

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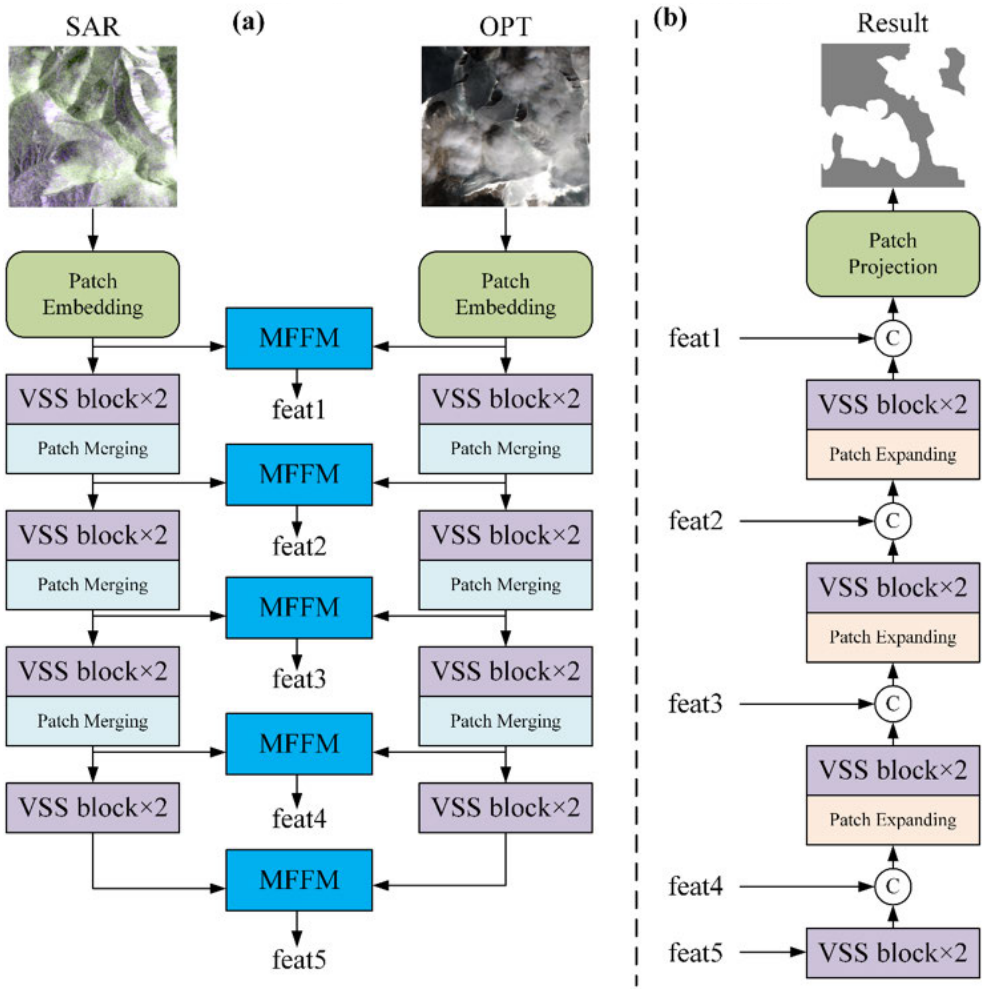


Fig. 3. Architecture of SAR-OPT-CDNet. (a) Encoder. (b) Decoder.

global modeling, achieving robust performance across challenging scenes, such as thin clouds and high-reflectance surfaces.

B. Multimodal Feature Fusion Module

The MFFM aims to integrate complementary information from optical and SAR modalities while explicitly modeling their differences in cloud–surface interactions. The module consists of three components: 1) joint feature extraction; 2) differential feature modeling; and 3) attention-guided fusion, as illustrated in Fig. 4.

1) Joint Feature Extraction: Given the optical feature map F_{opt} and the SAR feature map F_{sar} at the same scale, joint representations are first computed by channel-wise concatenation followed by a convolution layer:

$$F_{\text{joint}} = \text{Conv}\left(concat\left(F_{opt}, F_{sar}\right)\right)$$

where the convolution integrates modality-combined semantics and enhances features shared across the two domains, such as cloud thickness and large-scale texture patterns.

2) Differential Feature Modeling: In multimodal cloud detection, SAR often preserves structural information beneath cloud layers, while optical features capture radiometric changes caused by cloud thickness. Their discrepancies are therefore informative, especially for thin-cloud and boundary regions. To capture this complementary behavior, a differential representation is computed as

$$F_{\text{diff}} = \mid F_{opt} - F_{sar} \mid .$$

This operation highlights modality-inconsistent regions, effectively enhancing cloud edges and suppressing false alarms over high-reflectance surfaces where optical and SAR responses differ significantly.

A convolution layer is then applied to refine the differential representation

$$F'_{\text{diff}} = \text{Conv}\left(F_{\text{diff}}\right).$$

3) Attention-Guided Fusion: To adaptively emphasize informative channels and spatial structures, MFFM incorporates both channel attention (CAM) and spatial attention (SAM). Given the fused intermediate feature

$$F_c = F_{\text{joint}} + F'_{\text{diff}}$$

channel-wise attention is computed as

$$W_c = \sigma\left(\text{MLP}\left(\text{GAP}\left(F_c\right)\right)\right)$$

where GAP denotes global average pooling and σ denotes the sigmoid function. Spatial attention is obtained by

$$W_s = \sigma\left(\text{Conv}_{1\times 1}\left(\left[\text{AvgPool}\left(F_c\right)\text{MaxPool}\left(F_c\right)\right]\right)\right).$$

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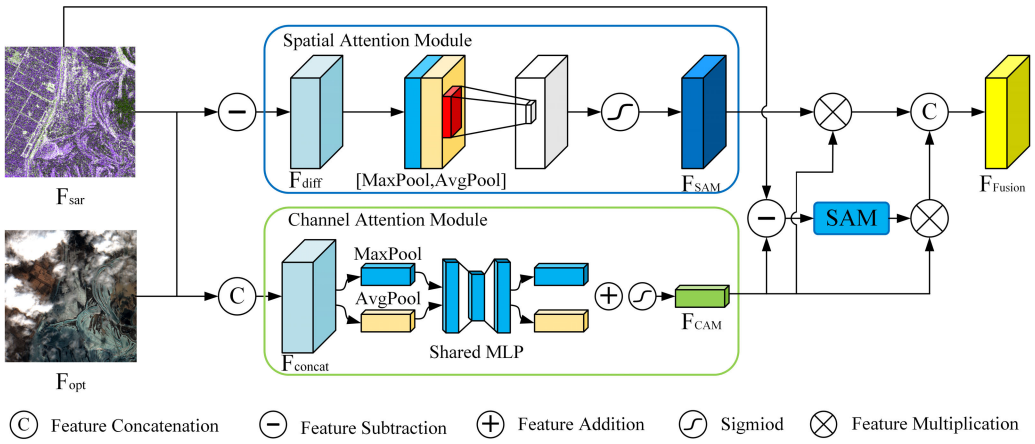


Fig. 4. Architecture of MFFM.

The final fused representation becomes

$$F_{\text{fused}} = F_c \odot W_c \odot W_s$$

enabling the network to enhance cloud-related features while suppressing redundant or noisy modal responses.

4) Contribution of MFFM: Through joint/differential modeling and attention-guided refinement, MFFM effectively integrates cross-modal cues and strengthens the discrimination of thin clouds, cloud boundaries, and high-reflectance surface regions. This design surpasses naive fusion operations, such as concatenation or simple addition, by explicitly leveraging the complementary nature of SAR and optical modalities.

C. VSS Block

To enhance long-range dependency modeling within the fused multimodal features, SAR-OPT-CDNet incorporates a VSS Block, a lightweight state-space module inspired by the recent Mamba architecture. Conventional convolutions have limited receptive fields and struggle to capture large-scale cloud structures, while transformers are computationally expensive for high-resolution remote sensing imagery. The VSS Block provides an efficient alternative by modeling spatial dependencies through selective scanning along multiple directions.

1) State-Space Modeling: The architecture of the VSS Block is shown in Fig. 5. Given an input feature map $F \in \mathbb{R}^{H\times W\times C}$, VSS treats each spatial sequence as a 1-D signal and processes it using a simplified state-space formulation

$$h_t = Ah_{t-1} + Bx_t$$
$$y_t = Ch_t$$

where

x_t —the sequentially scanned features;
h_t —the hidden state;
y_t —output sequence;
A, B, C —learnable parameters controlling the propagation of contextual information.

This formulation enables efficient global information aggregation while maintaining linear computational complexity with respect to sequence length.

2) Selective Scanning: To capture contextual cues from both cloud interiors and boundary regions, VSS applies bidirectional selective scanning

$$F_{\rightarrow}, F_{\leftarrow} = \text{SS2D}(F)$$

where SS2D performs forward and backward scanning along height and width directions, generating four directional feature streams. These streams encode:

The SS2D scan expansion and scan merging process is illustrated in Fig. 6.

a) vertical transitions (useful for elongated cloud patterns);
b) horizontal transitions (useful for cloud boundaries and thin clouds);
c) asymmetric responses between SAR and optical features.

The directional outputs are aggregated as

$$F_{\text{scan}} = \text{Concat}(F_{\rightarrow}, F_{\leftarrow})$$
$$F_{\text{vss}} = \text{Conv}_{1\times 1}(F_{\text{scan}}).$$

3) Gated Feature Selection: Following Mamba-like gating mechanisms, VSS applies content-dependent gating to regulate information flow

$$G = \sigma(\text{Conv}(F)), F_{\text{out}} = G \odot F_{\text{vss}}$$

where the gate G adaptively suppresses noise and emphasizes spatial structures relevant to cloud detection.

4) Effectiveness in Cloud Detection: The VSS Block benefits cloud detection in several ways.

a) *Thin Clouds:* Selective scanning captures weak and elongated structures that are easily missed by CNNs.
b) *Cloud Boundaries:* Bidirectional context enhances contrast transitions and improves segmentation sharpness.
c) *Multimodal Consistency:* SAR is more stable in high-reflectance regions; VSS propagates these cues globally to reduce optical-only false alarms.

Thus, VSS provides efficient large-scale spatial reasoning without the high cost of transformer-based models.

By incorporating the VSS block and SS2D module, SAR-OPT-CDNet achieves the following advantages.

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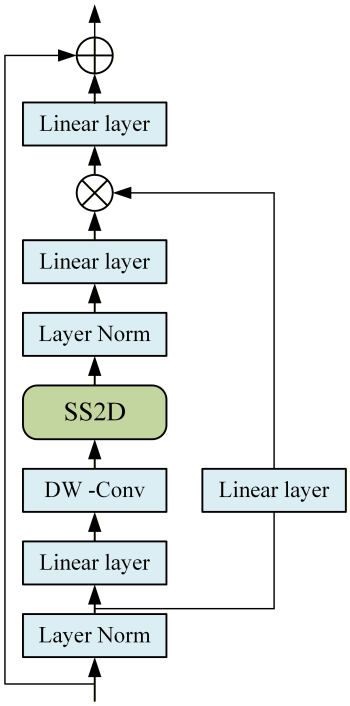


Fig. 5. Architecture of VSS Block.

1) *Multidirectional Feature Capture:* Through Scan Expanding and Scan Merging, the model fully leverages the multidirectional information in the input data, significantly enhancing the diversity and robustness of the features.
2) *Dynamic Selection Mechanism:* The S6 module dynamically adjusts the attention allocation mechanism, effectively distinguishing and retaining important information while filtering out irrelevant features, thereby improving the model's feature representation capability.
3) *Lightweight Design:* By combining depthwise separable convolution and the dynamic scanning mechanism, the VSS block not only enhances feature extraction performance but also significantly reduces computational complexity.

Different from conventional convolutional operations that rely on fixed local receptive fields, the VSS Block employs selective multidirectional scanning to aggregate long-range contextual information with linear computational complexity. This property is particularly suitable for cloud detection, where cloud structures often exhibit elongated shapes, irregular boundaries, and large spatial extents that are difficult to capture using standard convolutions or small-kernel dilated operations.

Compared with transformer-based self-attention, the VSS Block avoids quadratic complexity with respect to spatial resolution, making it more scalable to high-resolution remote sensing imagery. Meanwhile, its directional scanning mechanism explicitly models spatial continuity along multiple orientations, which aligns well with the geometric characteristics of thin clouds and cloud boundaries. Therefore, the VSS Block provides an efficient and task-adaptive solution for long-range context modeling in cloud detection, rather than serving as a generic architectural replacement.

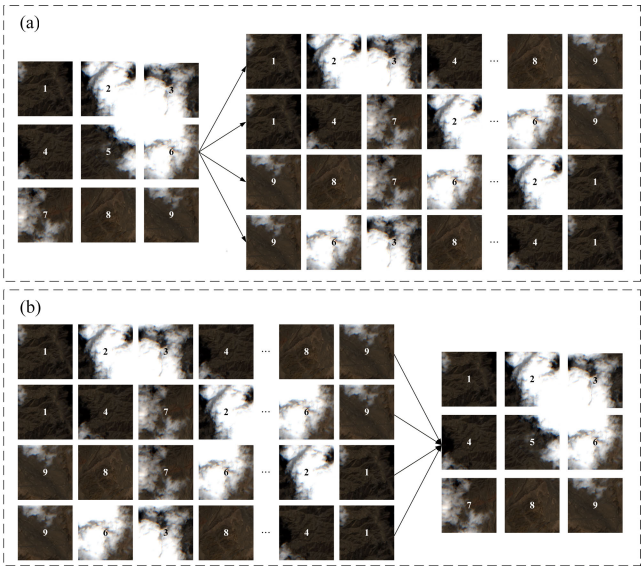


Fig. 6. Architecture of SS2D. (a) Scan expand. (b) Scan merge.

V. Experimental Results

A. Experimental Setting

The experiments in this study were conducted using the NUAA-SAR-OPT-CD dataset, with input images sized at 512×512 pixels. Python 3.10 and PyTorch 2.1.0 were used for network development and experimentation. The Adam optimizer was employed for training, with the maximum learning rate set at 0.0001. During training, the learning rate was progressively reduced to a minimum of 0.000001 once convergence was achieved. The batch size was set to 4, and the number of epochs was set to 100. We used the cross-entropy loss function as the primary criterion for model optimization. Cross-entropy loss is commonly used in both multiclass and binary classification tasks to quantify the difference between predicted probability distributions and true labels. It is defined as

$$L = -\frac{1}{w \times h} \sum_{i=1}^{w} \sum_{j=1}^{h} \left(y_{i,j} \ln \bar{y}_{i,j} + \left(1 - y_{i,j}\right) \ln \left(1 - \bar{y}_{i,j}\right) \right)$$

where $y_{i,j}$ is the reference cloud mask, $\bar{y}_{i,j}$ is the predicted mask. w and h are the width and height of the reference mask.

B. Accuracy Metrics

In order to evaluate the performance of cloud detection methods quantitatively, four accuracy metrics were selected: 1) overall accuracy (OA); 2) precision or producer's accuracy (PA); 3) recall or user's accuracy (UA); 4) F1-score. The formulas of accuracy metrics are as follows:

$$OA = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

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TABLE IV
AVERAGE ACCURACY INDICATORS OF DIFFERENT METHODS

	OA	Precision	Recall	F1-score
Fmask	91.98%	40.45%	**97.79%**	0.5723
RSNet	80.25%	91.70%	86.53%	0.8904
Boundary Net	99.01%	92.61%	89.09%	0.9081
ECD-Net	79.69%	94.61%	83.49%	0.8870
RD-Unet	56.71%	62.58%	85.81%	0.7238
SAR-OPT-CDNet	**99.13%**	**95.25%**	88.56%	**0.9178**



Fig. 7. Cloud detection results of different methods. (a) OPT. (b) SAR. (c) Label. (d) Fmask. (e) RS-Net. (f) Boundary Net. (g) ECD-Net. (h) RD-Unet. (i) CERMF-Net. (j) CFFormer. (k) OPT-SAR-CDNet.

$$Recall = \frac{TP}{TP + FN}$$
$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

C. Experimental Results

Table IV presents the quantitative comparison of different cloud detection methods in terms of OA, precision, recall, and F1-score. Among the optical-based baselines, BoundaryNet achieves relatively balanced performance across evaluation metrics, whereas Fmask attains the highest recall but suffers from substantial false detections, resulting in low precision and F1-score. RS-Net, ECD-Net, and RD-Unet improve precision to varying degrees; however, they still struggle to achieve a robust trade-off between omission and commission errors.

In contrast, the proposed SAR-OPT-CDNet achieves the highest OA and F1-score among all cloud detection methods, indicating superior overall detection performance. Although its recall is slightly lower than that of Fmask, the significantly higher precision demonstrates that SAR-OPT-CDNet effectively suppresses false alarms, which is particularly important for cloud detection over high-reflectance surfaces. This behavior is consistent with the qualitative comparisons shown in Figs. 7 and 8.

In addition, two representative optical–SAR fusion methods, CERMF-Net [40] and CFFormer [41], are included for qualitative comparison. Since these methods are originally designed for general optical–SAR image fusion or interpretation tasks rather than pixel-level cloud detection, their outputs are not directly optimized for cloud mask generation. Therefore, quantitative cloud detection metrics are not reported for these methods. Instead, their performance is analyzed qualitatively through visual comparisons, which reveal that although SAR information improves structural consistency to some extent, these methods still exhibit fragmented detections and residual commission errors in thin-cloud and high-reflectance scenarios compared with the proposed SAR-OPT-CDNet.

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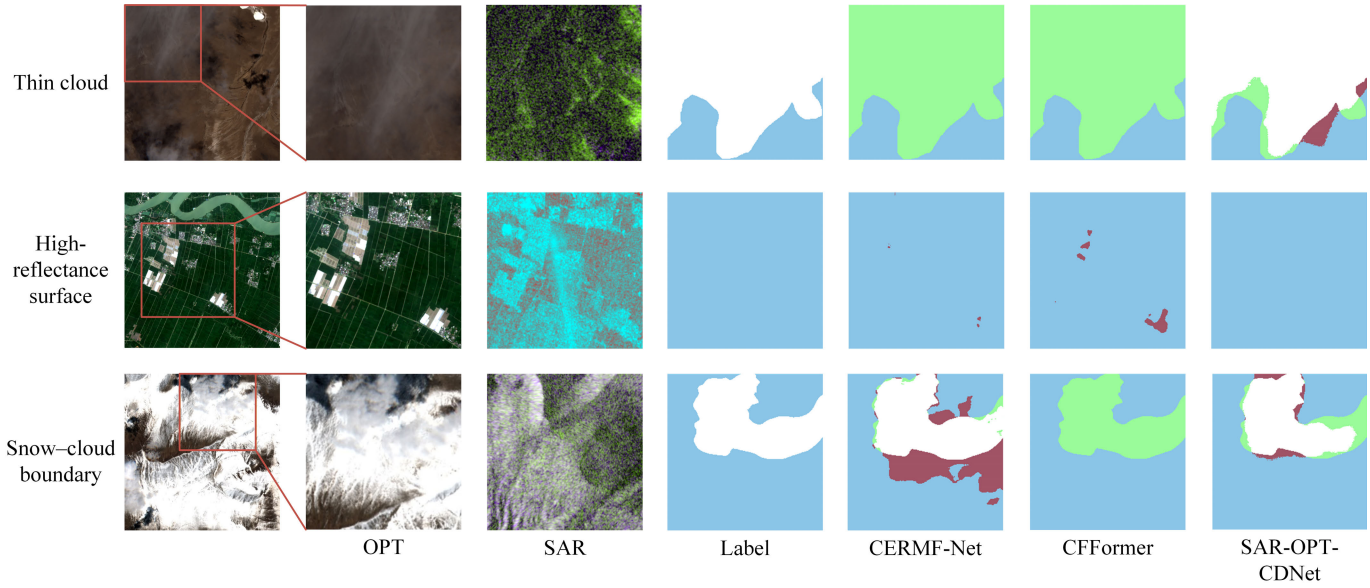


Fig. 8. Cloud detection results of optical–SAR fusion methods in challenging regions.

Fig. 7 presents a comprehensive visual comparison of cloud detection results obtained by different methods under diverse surface and atmospheric conditions.

In the first row, thin cloud scenes are illustrated, where cloud reflectance is weak and spectral contrast is low. In this scenario, Fmask fails to detect most thin clouds, while RS-Net and BoundaryNet partially detect cloud regions but suffer from evident omission errors. ECD-Net and RD-Unet produce additional false detections. In contrast, the proposed method effectively compensates for the weak optical signals by incorporating SAR polarization information, enabling reliable detection of translucent clouds and achieving the most complete cloud masks among all compared methods.

The second and third rows show detection results over high-reflectance surface regions. Fmask, which relies primarily on spectral thresholding, produces relatively few false detections in these scenes. However, due to the similarity between the color and texture characteristics of clouds and bright surfaces, deep learning-based optical methods, such as RS-Net, BoundaryNet, ECD-Net, and RD-Unet exhibit noticeable commission errors. By jointly exploiting optical and SAR information, the proposed method effectively suppresses false alarms over bright surfaces and achieves more accurate surface–cloud discrimination.

The fourth and fifth rows correspond to snow and snow–cloud mixed scenarios, which represent particularly challenging cases for cloud detection. The coexistence of snow-covered terrain and clouds leads to strong feature ambiguity, causing most optical-based methods to misclassify clouds as snow or vice versa. Although Fmask detects most cloud regions, it still exhibits missed detections. RS-Net and BoundaryNet fail to correctly identify clouds over snow-covered mountains, while ECD-Net and RD-Unet generate false positives. In contrast, the proposed method demonstrates superior robustness by accurately detecting clouds above snow-covered terrain while minimizing false detections.

Overall, Fig. 7 demonstrates that the proposed method consistently achieves high-precision cloud detection across different challenging scenarios, including thin clouds, bright surfaces, and snow-covered regions, while effectively reducing both omission and commission errors compared with existing methods.

Fig. 8 further analyzes the effectiveness of optical–SAR fusion strategies by providing zoomed-in comparisons over representative challenging regions.

The first row focuses on thin cloud regions with extremely weak spectral signatures. Although both CERMF-Net and CFFormer benefit from the inclusion of SAR information, their detection results remain fragmented, with incomplete cloud structures and residual omission errors. In contrast, the proposed SAR-OPT-CDNet preserves thin cloud continuity more effectively, yielding more complete and coherent cloud masks.

The second row illustrates cloud detection results over high-reflectance surface regions. Due to the strong similarity between bright surfaces and clouds in optical imagery, CERMF-Net and CFFormer still exhibit commission errors, despite incorporating SAR features. By explicitly modeling joint and differential cross-modal features, SAR-OPT-CDNet suppresses false detections over bright surfaces and achieves clearer separation between clouds and underlying land covers.

The third row shows snow–cloud boundary regions, where mixed pixels and complex terrain further complicate cloud detection. Existing fusion methods produce blurred boundaries and misclassifications between snow and cloud regions. In contrast, SAR-OPT-CDNet generates sharper cloud boundaries and more accurate spatial delineation, benefiting from efficient multidirectional context modeling enabled by the VSS Block.

These results indicate that, beyond simply incorporating SAR data, explicitly disentangling and fusing complementary cross-modal information is critical for improving cloud detection performance in challenging scenarios.

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TABLE V
COMPARISON OF PREDICTION ACCURACY OF DIFFERENT METHODS. THE BEST VALUES ARE MARKED IN BOLD

		VSS	SAR	Fusion	OA	Precision	Recall	F1-score
Exp1	Unet	×	×	×	62.93%	81.80%	73.18%	0.7725
Exp2	Unet+SAR	×	√	Cat	73.51%	93.35%	77.57%	0.8473
Exp3	Unet+SAR+CBAM	×	√	CBAM	71.70%	86.24%	80.96%	0.8352
Exp4	Unet+SAR+MFFM	×	√	MFFM	74.94%	95.20%	77.70%	0.8556
Exp5	Unet+VSS	√	×	×	98.83%	93.14%	84.93%	0.8885
Exp6	Unet+VSS+SAR	√	√	Cat	98.91%	93.48%	86.24%	0.8971
Exp7	SAR-OPT-CDNet	√	√	MFFM	**99.13%**	**95.25%**	**88.56%**	**0.9178**

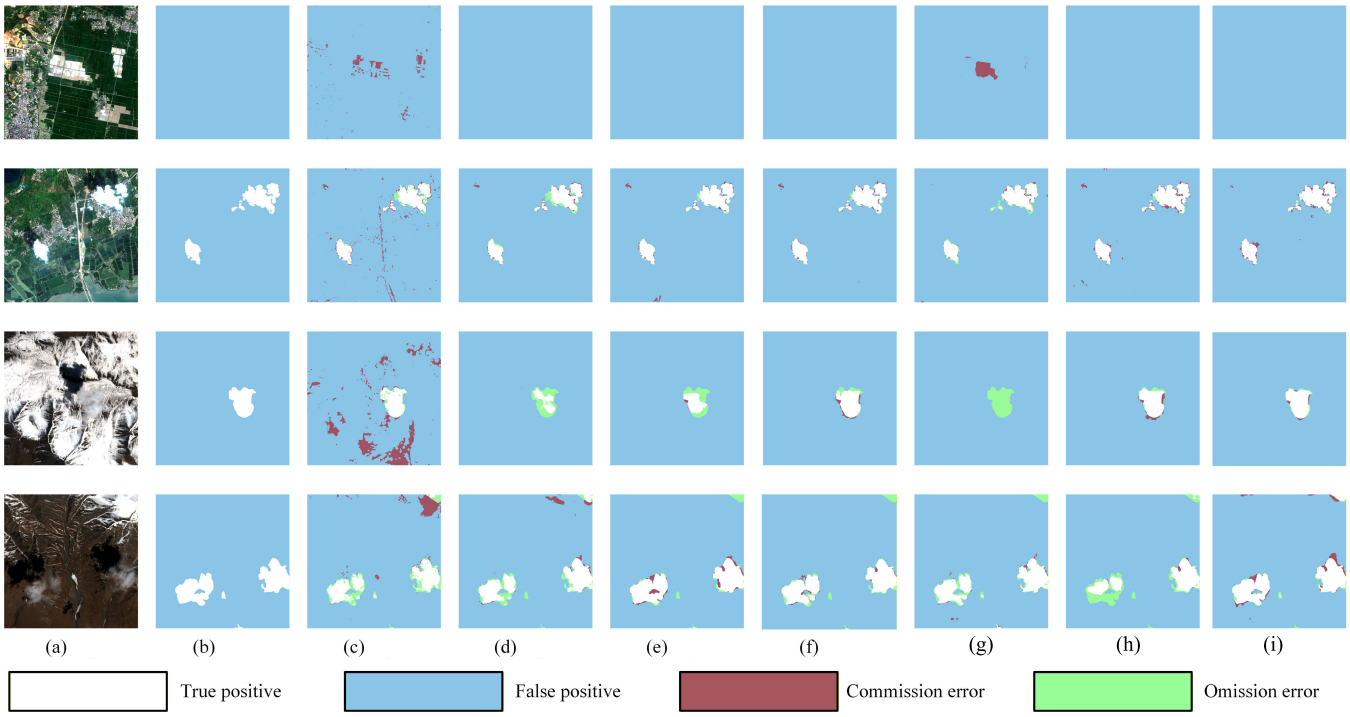


Fig. 9. Comparison of ablation experimental results. (a) img. (b) label. (c) Exp1. (d) Exp2. (e) Exp3. (f) Exp4. (g) Exp5. (h) Exp6. (i) OPT-SAR-CDNet.

D. Ablation Experiment

To assess the contribution of individual components in SAR-OPT-CDNet, a series of ablation experiments were conducted based on a classical encoder–decoder UNet architecture. As summarized in Table V, SAR feature integration, the VSS Block, and different fusion strategies were progressively introduced to the baseline model. Representative qualitative results are illustrated in Fig. 9.

The results indicate that incorporating SAR information consistently improves cloud detection performance by reducing misclassification over high-reflectance surfaces. The introduction of the VSS Block further enhances contextual feature modeling, leading to improved recall and more coherent cloud structures, particularly in thin-cloud and boundary regions. Moreover, comparisons between different fusion strategies show that generic attention-based fusion (e.g., CBAM) provides limited gains, whereas the proposed MFFM achieves more substantial improvements by explicitly modeling joint and differential cross-modal features. By jointly integrating SAR features, efficient long-range contextual modeling, and difference-aware multimodal fusion, the full SAR-OPT-CDNet achieves the best overall performance across all evaluation metrics.

While this study primarily focuses on discriminating clouds from high-reflectance surfaces, the potential of SAR for cloud-shadow detection is a promising direction. Unlike optical imagery where shadows cause a sharp drop in radiance, SAR is insensitive to solar illumination. However, SAR could theoretically assist in identifying "geometric shadows" by detecting local backscattering anomalies. Due to the current dataset's focus on cloud masks and the distinct physical scattering mechanisms of shadows, we designate cloud-shadow integration as a priority for our future work.

E. Operational Efficiency

Table VI presents a quantitative comparison of the operational efficiency of different cloud detection methods. The evaluation includes processing speed (measured in images per second), floating-point operations (FLOPs), and model size, which together reflect the computational cost and deployment feasibility of each method.

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TABLE VI
OPERATIONAL EFFICIENCY OF DIFFERENT METHODS

	Efficiency (images/s)	Flops	Model Size (MB)
Fmask	2.19	-	-
RS-Net	26.98	37.91	29.62
BoundaryNet	4.02	510.01	1018.05
ECD-Net	26.53	193.37	75.4
RD-Unet	8.65	218.95	120
CERMF-Net	18.17	236.83	154
CFFormer	13.20	280.48	568
SAR-OPT-CDNet	10.28	206.21	110.10

Traditional methods, such as Fmask exhibit very low processing speed but negligible computational overhead, as they rely primarily on rule-based operations. Among deep learning-based optical methods, RS-Net achieves the highest processing speed with relatively low FLOPs and a compact model size, while BoundaryNet attains competitive accuracy at the expense of substantially increased computational complexity and memory consumption.

Compared with optical-only deep learning models, multimodal fusion methods generally introduce additional computational cost due to the incorporation of SAR features. CERMF-Net and CFFormer achieve moderate processing speeds but require larger model sizes and higher FLOPs, reflecting the overhead of complex fusion and attention mechanisms.

The proposed SAR-OPT-CDNet achieves a favorable balance between detection performance and operational efficiency. Although its processing speed is lower than that of lightweight optical-only models, SAR-OPT-CDNet maintains a moderate computational complexity and model size while significantly improving cloud detection robustness in challenging scenarios. This balance makes the proposed method suitable for practical remote sensing applications where both accuracy and computational efficiency are critical considerations.

VI. CONCLUSION

This article presents SAR-OPT-CDNet, a robust multimodal fusion framework designed to enhance cloud detection by synergistically leveraging optical and SAR imagery. To address the spectral ambiguity between clouds and high-reflectance surfaces, we developed the MFFM, which effectively captures both shared representations and cross-modal discrepancies. By incorporating Mamba-inspired VSS blocks, the network achieves extensive receptive fields and multidirectional contextual modeling with linear computational complexity, significantly improving the segmentation of thin clouds and complex cloud boundaries.

Extensive experiments on the NUAA-SAR-OPT-CD dataset demonstrate that the proposed method consistently outperforms state-of-the-art approaches, achieving an OA of 99.13% and an F1-score of 0.9178. The results confirm that integrating SAR's structural information effectively mitigates false alarms in bright terrestrial regions and omission errors in semi-transparent cloud zones. Furthermore, the framework exhibits strong robustness to minor temporal misalignments between sensors, making it highly practical for real-world remote sensing applications. Future work will focus on expanding the model's generalizability to diverse satellite sensors and exploring self-supervised learning techniques to reduce the dependency on large-scale pixel-level annotations.

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